

Utkarsh Yashvardhan

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Education

Georgia Institute of Technology
MS Computer Science, 3.83/4

Atlanta, Georgia
Aug. 2024 – Present

BITS Pilani
M.Sc. Mathematics & B.E. Computer Science (Dual Degree), 8.47/10

Pilani, Rajasthan
Aug. 2019 – Jun. 2024

Experience

Academic.....

Work Integrated Learning Programmes (WILP) **BITS Pilani**
Teaching Assistant - Command Line Interface & Scripting *May 2024 – Jul. 2024*

Lead Teaching Assistant - Object Oriented Programming *Feb. 2024 – Apr. 2024*

Department of Computer Science & Information Systems **BITS Pilani**
Teaching Assistant - Data Structures & Algorithms *Jan. 2024 – May 2024*

Teaching Assistant - Computer Programming *Aug. 2023 – Dec. 2023*

BITS IOT Lab **BITS Pilani**
Research Assistant *May. 2023 – Jul. 2023*

○ Worked on applications of **Generative Artificial Intelligence (GAI)** in the security domain.

Open Source.....

JuliaTopOpt/SimpleTopOpt.jl **Remote**
Collaborator *Feb. 2024 – Jun. 2024*

○ **Translated** complex **MATLAB**-based **topology optimization** models (3D, stress, and buckling) into **Julia** using established academic literature.

○ **Benchmarked** the **computational efficiency** of the newly developed Julia implementations against their MATLAB counterparts.

Publications

2024: Generative AI for Cyber Security: Analyzing the Potential of ChatGPT, DALL-E and Other Models for Enhancing the Security Space | Siva Sai, [Utkarsh Yashvardhan](#), Vinay Chamola, and Biplab Sikdar, IEEE Access, 2024

Master thesis (BITS Pilani)

Title: *Bokeh Rendering using Federated Learning*

Supervisor: Dr Pratik Narang

Co-supervisors: Dr Jagat Sesh Challa, Dr Rakhi

○ Engineered **privacy-preserving Federated Learning (FL)** architectures from centralized vision models (**BRVIT & Stacked DMSHN**) using the **Flower** framework.

○ Integrated a **ViT-Large backbone** into BRVIT, elevating the centralized baseline to a **peak PSNR of 24.89**.

○ Achieved **24.42 PSNR** and **0.893 SSIM** in a **3-client decentralized simulation** on the **EBB! dataset**, matching baseline accuracy while **protecting user data**.

Projects

Georgia Institute of Technology

Atlanta, Georgia

Machine Learning Algorithms & Analysis

Fall 2025

Python, PyTorch, Scikit-learn, mlrose-hiive, OpenAI Gym

- Built scalable **Neural Networks** and **SVMs**, optimizing weights via **Simulated Annealing** and **Randomized Hill Climbing**.
- Benchmarked **K-Means** and **GMMs** using **ARI/NMI** scoring, extracting latent features via **PCA, ICA, and RP**.
- Solved Markov Decision Processes (MDPs) by evaluating Model-Based (**Value/Policy Iteration**) and Model-Free (**Q-Learning, SARSA**) algorithms.

Network Science and Graph Machine Learning

Summer 2025

Python, NetworkX, PyTorch Geometric, Scikit-learn, EoN

- Evaluated empirical **small-world** networks and benchmarked **community detection** algorithms (**Louvain, K-Clique**) using **NMI** scoring.
- Simulated **SIS epidemic outbreaks** on human contact networks via **EoN** to analyze transmission dynamics.
- Engineered **Graph Attention Networks (GATs)** and label propagation models for graph-based **NLP** tasks.

Bayesian Estimation of Differential Equations

Fall 2024

Julia, Turing.jl, OrdinaryDiffEq.jl, Bayesian Inference, MCMC

- Simulated non-linear **SEIR** compartmental disease models using **OrdinaryDiffEq.jl**.
- Executed **Bayesian parameter estimation** on noisy data utilizing **Turing.jl** and the **NUTS MCMC** algorithm.
- Validated estimations through **data retrodiction**, generating posterior ensembles to **quantify uncertainty**.

Technical Skills

Languages: C/C++, Python, Julia, MATLAB, Java, C#, SQL

ML & Scientific: PyTorch, Flower, NumPy, Scikit-learn, Qiskit, Pymc, Networkx, Pandas, Matplotlib

Developer Tools: Git, VS Code, Jupyter Notebook, IntelliJ IDEA, Unity, LaTeX

Scholarship

BITS Pilani

Pilani, Rajasthan

Merit Scholarship (II Semester 2019-2020)

Feb. 2021

Got Merit Scholarship of about Rs 70,000 which is given to **top 2% performers** in the batch.

Leadership

Microsoft Learn Student Ambassadors

Competitive Coding SIG Co-lead

BITS Pilani Chapter

Aug. 2021 – Apr. 2022

References

Ref 1: Dr. Pratik Narang

Associate Professor, Department of Computer Science & Information Systems

BITS Pilani, Pilani Campus

Email: pratik.narang@pilani.bits-pilani.ac.in

Relationship: First-Degree Thesis Advisor

Ref 2: Dr. Siva Sai

Postdoctoral Researcher, Department of Computer Science

NUS, Singapore

Email: p20220063@pilani.bits-pilani.ac.in

Relationship: Co-author & Direct Research Mentor

Ref 3: Dr. Vinay Chamola

Associate Professor, Department of Electrical & Electronics Engineering

BITS Pilani, Pilani Campus

Email: vinay.chamola@pilani.bits-pilani.ac.in

Relationship: Research Advisor & Co-author